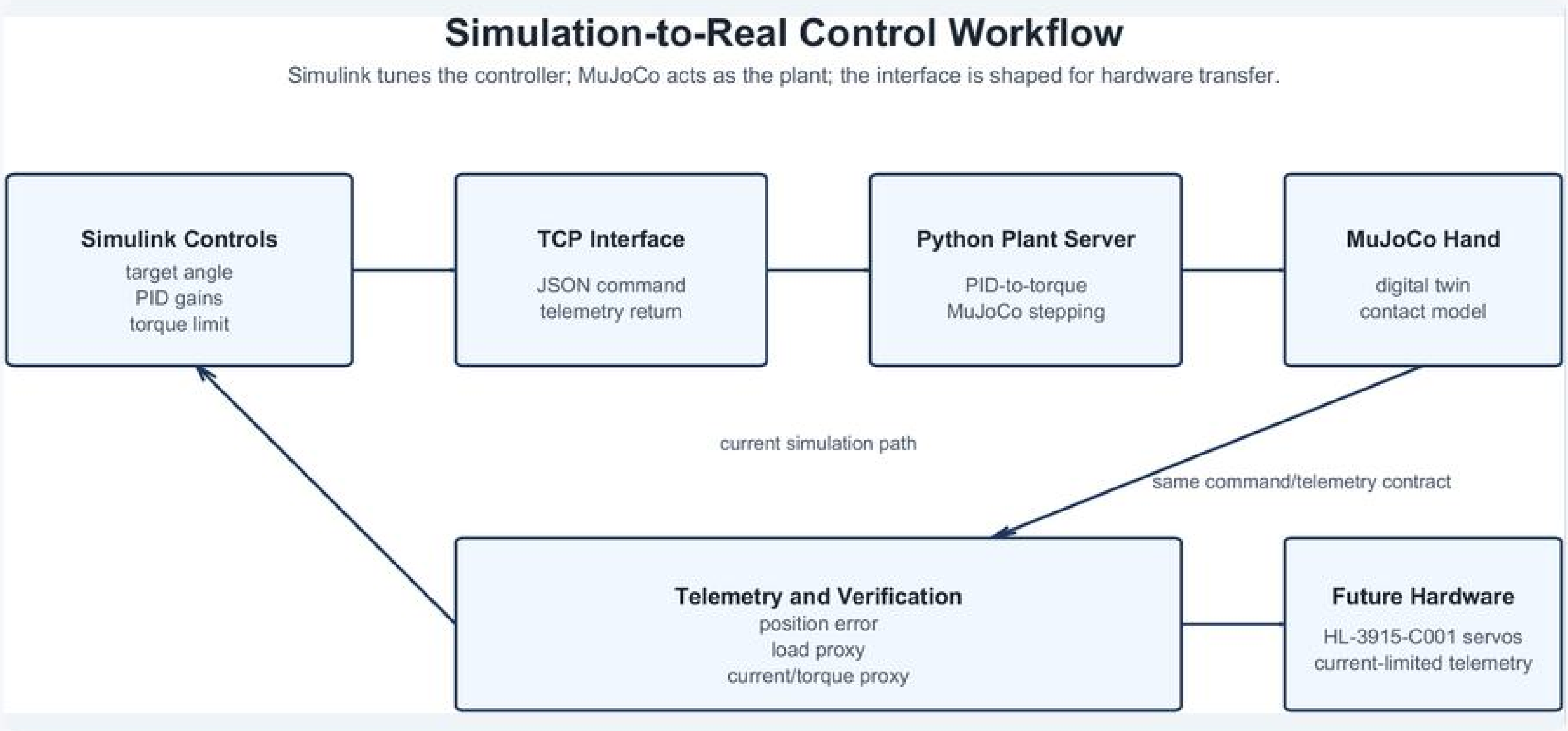
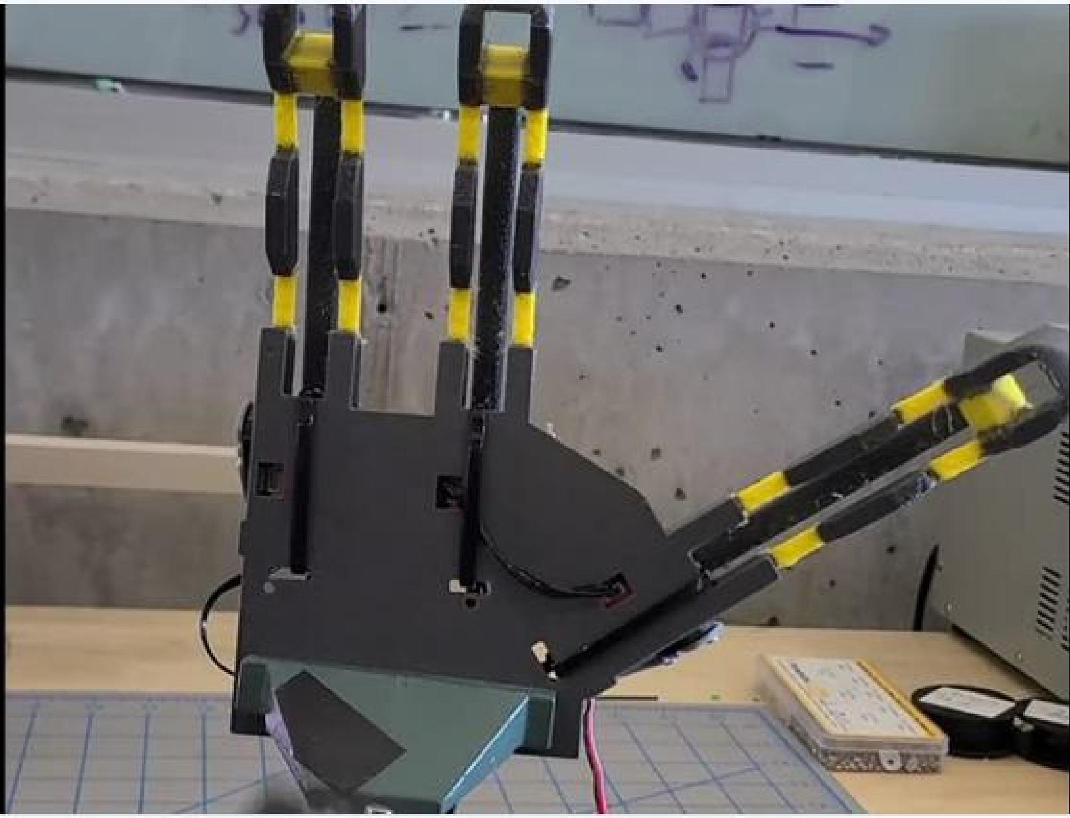



PORTFOLIO OVERVIEW

# Robotics R&D from mechanism concept to validated hardware

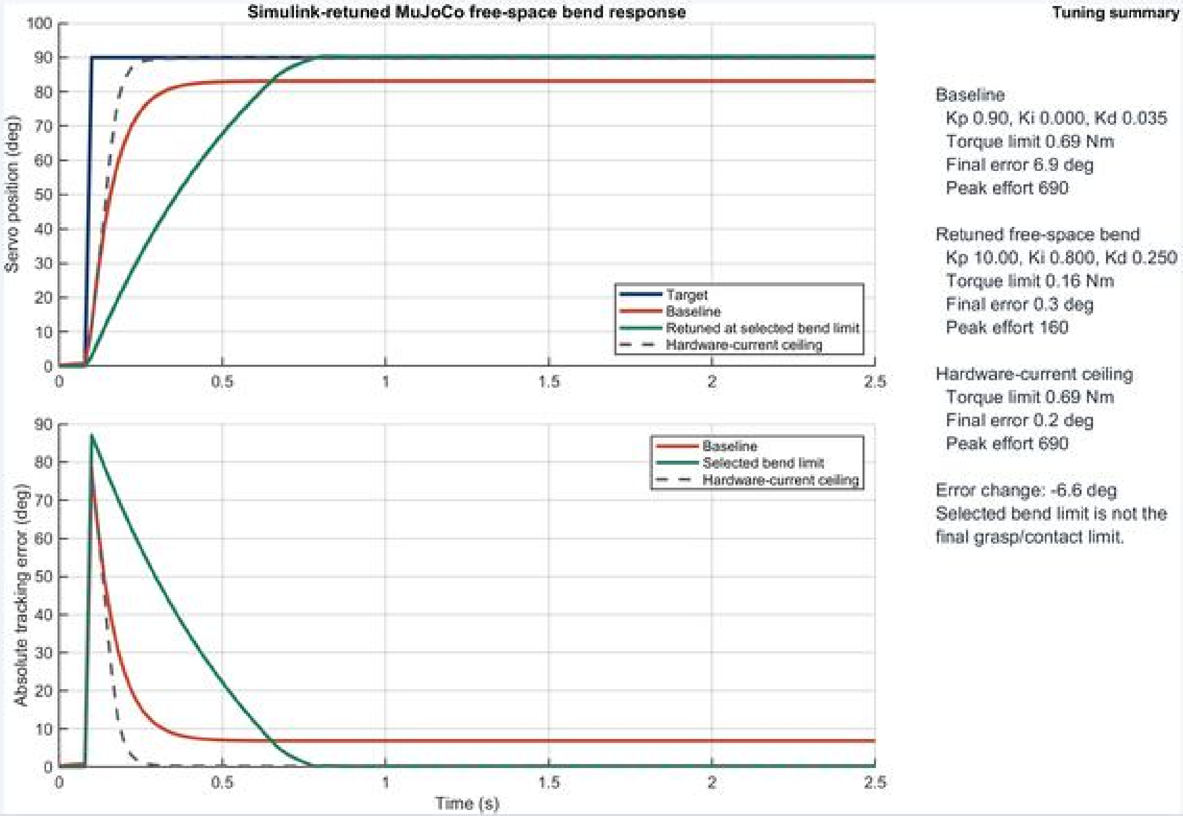
- PhD candidate finishing Summer 2026
- 8+ years robotics and mechatronics R&D
- CAD, controls, firmware, fabrication, and validation



COMPLIANT ROBOTIC HAND

# FDM-compatible hand with simulation-to-hardware controls

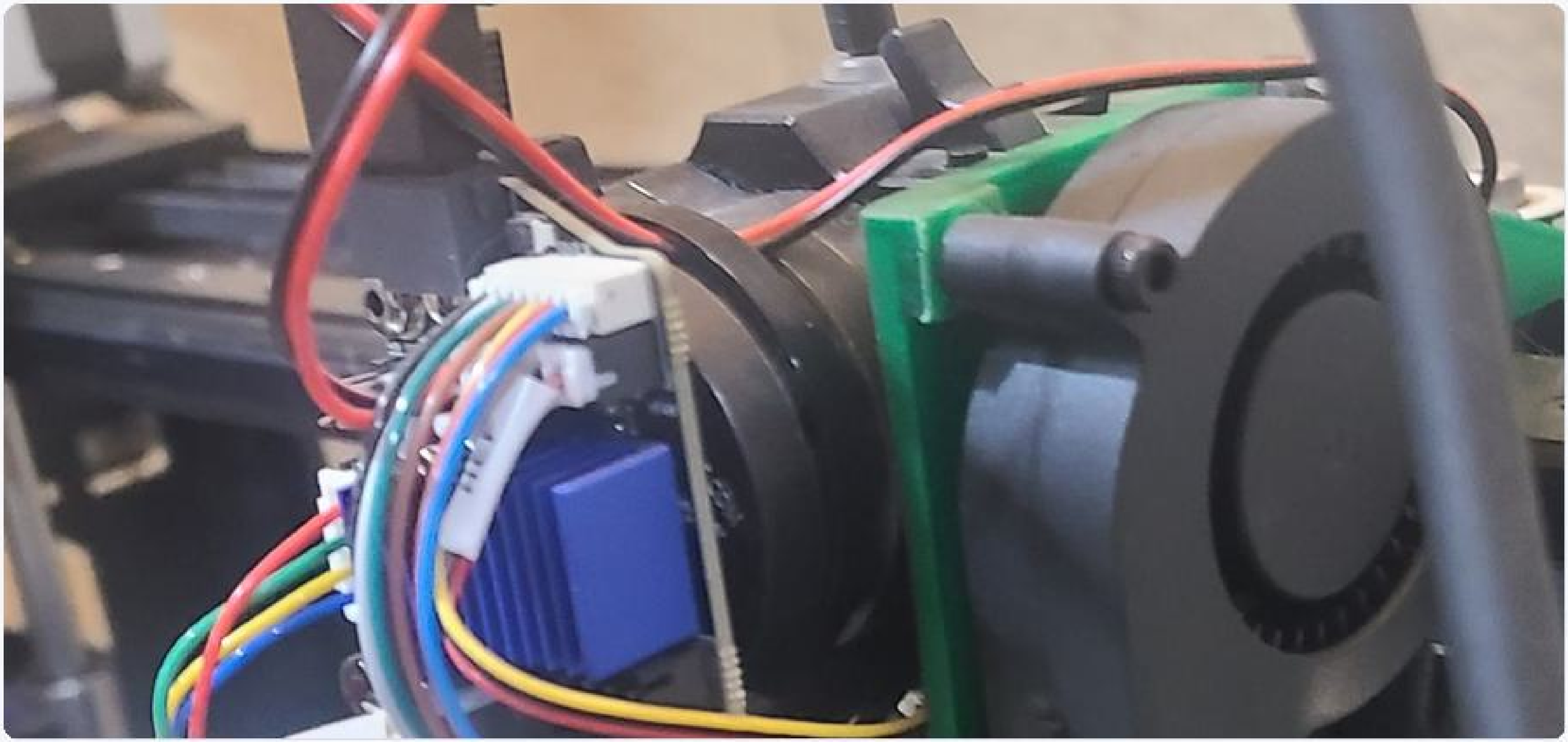
- Three-motor compliant mechanism
- Simulink/MuJoCo control workflow
- Python Feetech servo tooling and bench demos




### CONTROLS AND MODELING

# Simulink, MuJoCo, and motion-control evidence

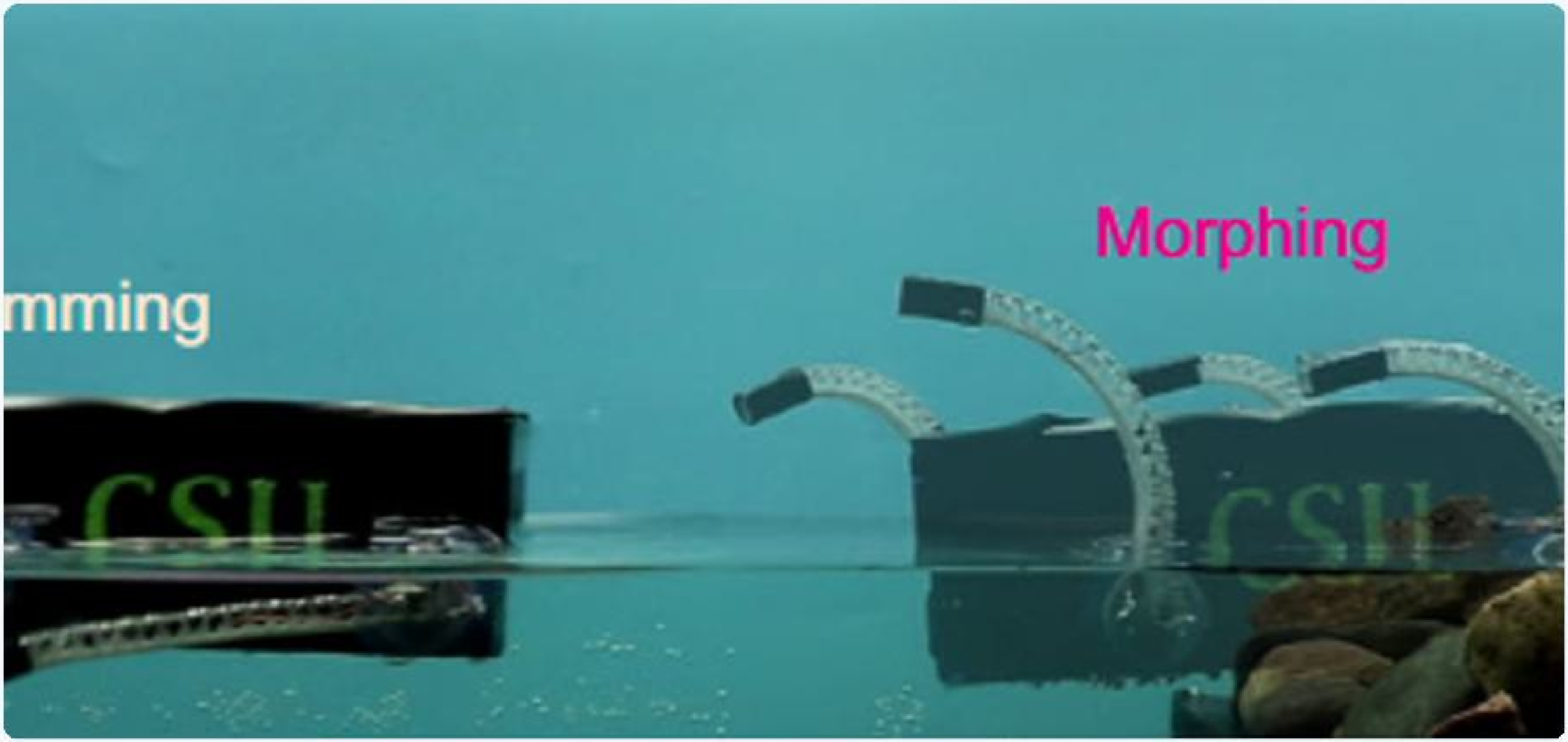
- Plant/controller separation
- PID tuning and torque/current limit sweeps
- Robot-learning and dynamics coursework artifacts



PRINTER RETROFIT

# Practical machine integration for additive manufacturing

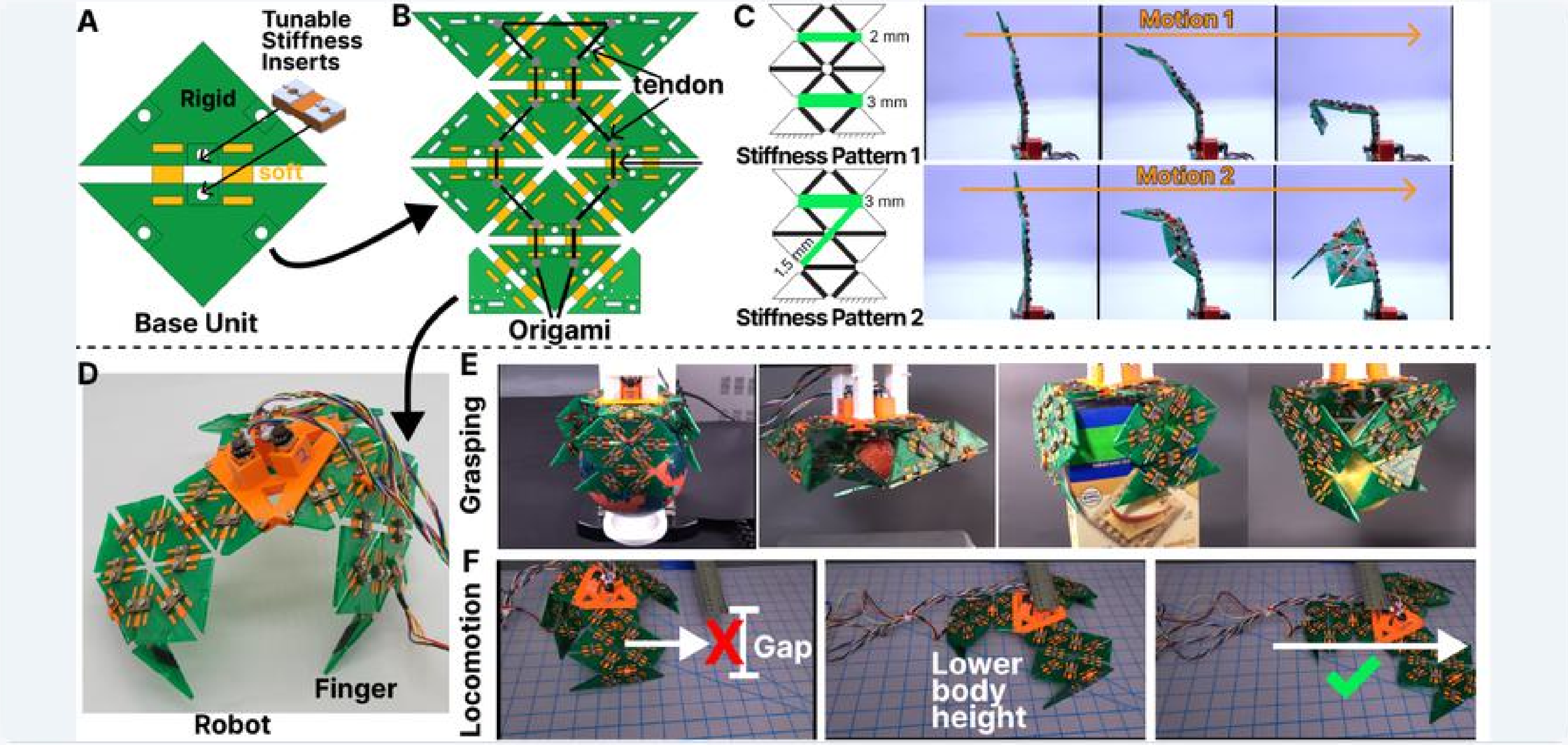
- Right-extruder Smart Orbiter-style retrofit
- CAD packaging and cooling duct design
- Custom Klipper startup G-code




SHAPE-MORPHING ROBOT

# Amphibious system integration and validation

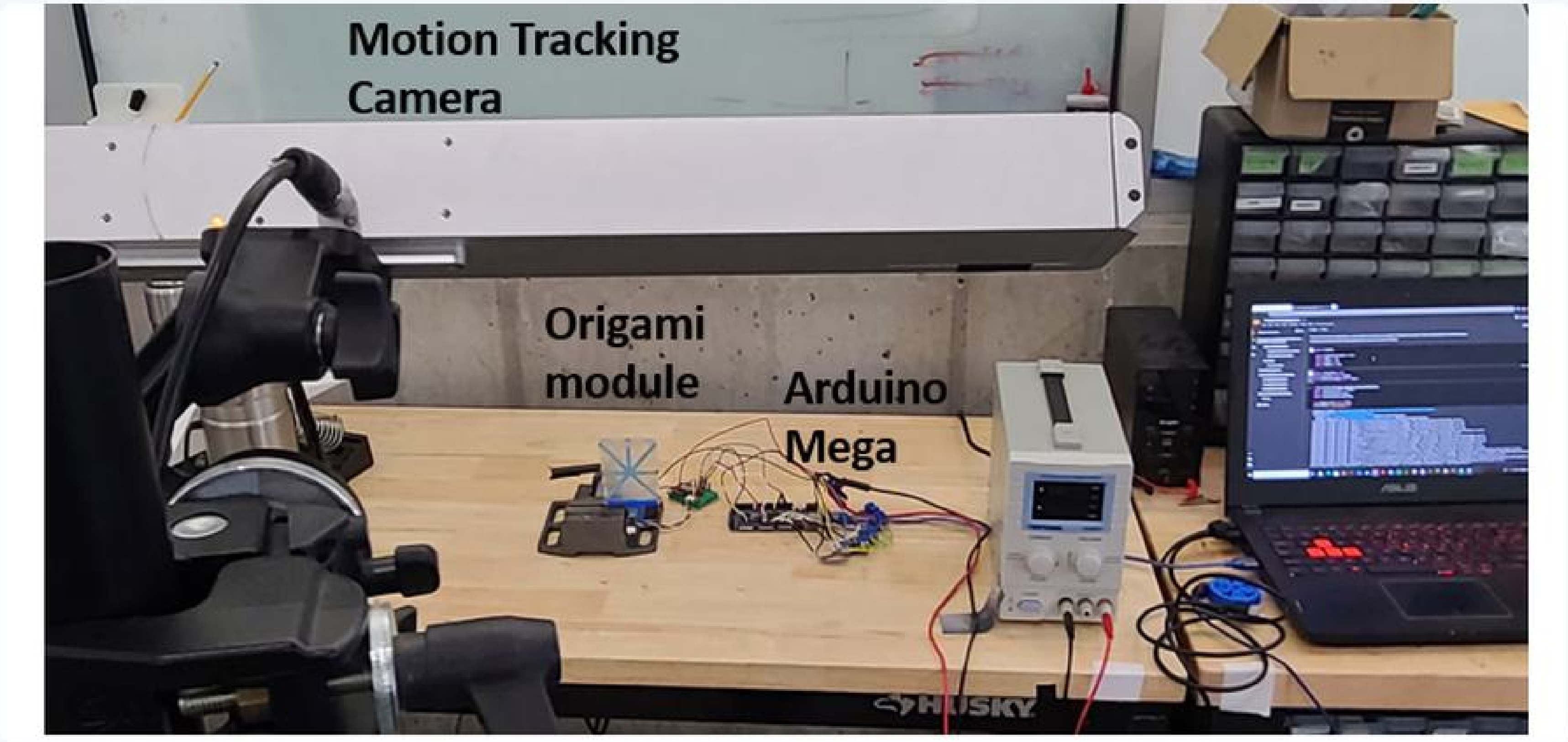
- Waterproof body and morphing limbs
- Encoded motors, BLE modes, thermal actuation
- Tank tests and outdoor land-water transitions




VARIABLE-STIFFNESS ROBOTS

# Origami platforms with active stiffness control

- SMP/PETG composites and active VSIs
- Thermistors, heaters, relay isolation, encoders
- Locomotion, gait switching, and adaptive grasping




EXPLORE THE WORK

# Portfolio, source evidence, and publications

- Full project pages: elishalerner.github.io
- Source files and control artifacts included
- Available for robotics R&D and mechatronics roles after PhD completion